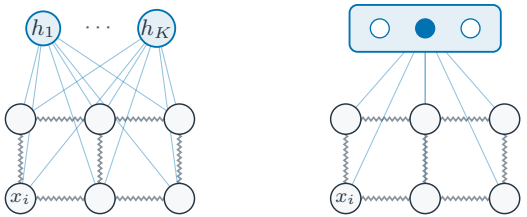
(a) Visible and global hidden units

Ising: K binary units
$h_j \in \{-1, +1\}$

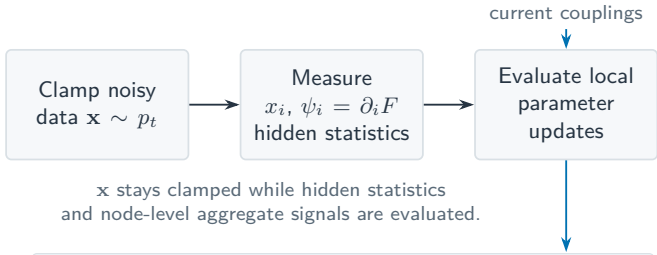
Potts: one M-state unit
one state active



Continuous visible oscillators; nearest-neighbour springs

Global hidden–visible couplings
Optional per-pixel Ising units are omitted.

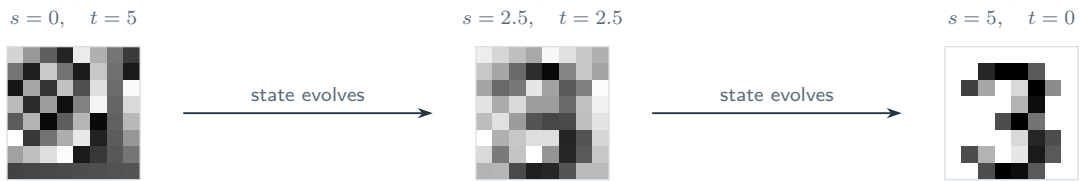
(b) Local training at each noise time t



Fast hidden equilibration; ψ_i is the mean clamp force.

(c) Generation by open-loop parameter replay

Prelearned protocol $\theta(5 - s)$: the same schedule for every generated sample



$$\mathrm{d}\mathbf{x} = \left[\mathbf{x} - 2\nabla F_{\theta(5-s)}(\mathbf{x})\right] \mathrm{d}s + \sqrt{2T}\,\mathrm{d}\mathbf{B}_s$$

Effective visible dynamics after averaging the fast hidden units

Schematic network subsets and illustrative image sequence; no measured or generated samples are shown.